

Self-Intersection-Aware Deformation Transfer for Garment Simulation Meshes

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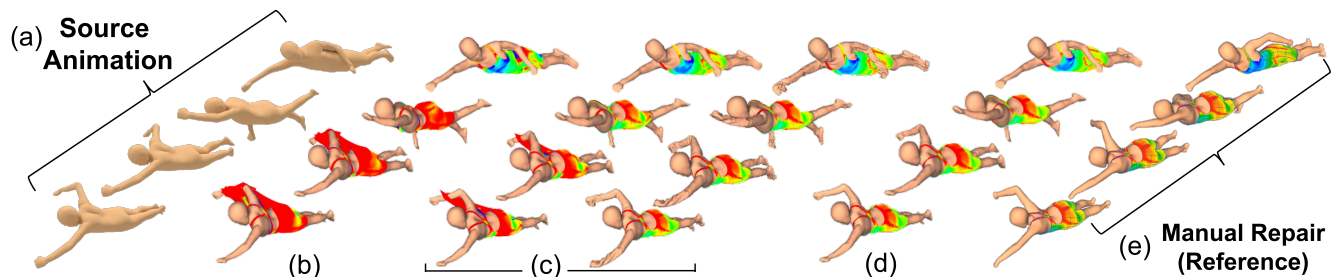


Figure 1: Garment simulation is performed on the mesh generated by DT (b), DT + *Animation Repair* [Fang et al. 2021] (c), SIDT (d) and DT with manual revision (e) by adjusting the rig of the source model (a). Our SIDT removes the self-intersections from the simulation, which is caused by collapsing cloth mesh when the right arm intersects the body mesh. Where, green represents neutral, while red and blue denote tensile strain and compressive strain of the prototype sportswear, respectively

ABSTRACT

Deformation Transfer is a well-known technique which copies deformations between two meshes to re-use animation. One drawback of existing methods is that they can introduce self-intersections on the resulting mesh, even when there is no self-intersections on the source mesh. We propose a novel deformation transfer technique that suppresses self-intersections on the resulting target mesh, called *Self-Intersection-aware Deformation Transfer* (SIDT). We demonstrate that SIDT helps to produce simulation friendly mesh for garment simulation.

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1 INTRODUCTION

Deformation Transfer, referred to as DT, is a well known technique for copying deformations from a source mesh to a different target mesh. Applications of DT include creating facial blend shapes, avatar reshaping, and whole body character animation [Roberts et al. 2021]. DT, and more recent variants, do not explicitly account for potential self-intersections. Such self-intersections are problematic when the resulting animation is used as input to other processes. For example, when the resulting animation is fed into a cloth simulator, undesirable artifacts will occur (c.f. mesh collapsing like top row of Fig. 1(b)). In this situation animators or modelers need to fix manually the resulting animation, or employ physics simulation to resolve self-intersections on the mesh as a post processing [Fang et al. 2021; Wong et al. 2018]. Unfortunately, these approaches generate additional costs in the production pipeline.

To address this problem, we propose a novel deformation transfer technique that suppresses self-intersections on the resulting target mesh. We call our technique *Self-Intersection-aware Deformation Transfer* (SIDT). Compared with *Animation Repair* [Fang et al. 2021], we demonstrate that our method is easy to implement but generates mesh animations of good quality. We illustrate our method with an application to garment simulation.

2 SELF-INTERSECTION-AWARE DEFORMATION TRANSFER

2.1 Deformation Transfer

DT transfers the *deformation gradients* of animations from a source mesh S onto the target mesh $\mathcal{T}$. Once the transfer has been applied, the deformation gradients $\mathbf{D}^{\tilde{\mathcal{T}}}$ for all faces F on the target mesh $\tilde{\mathcal{T}}$ will closely match those of their correspondingsource mesh faces $\mathbf{D}^S$. Specifying one or more positional constraint $\mathbf{p}^i$, DT finally may be formulated as a sparse linear system with positional constraints.

$$E_{dt} = \|\mathbf{A}\mathbf{x} - \mathbf{B}\|_2^2 \quad \text{s.t.} \quad \mathbf{x}_k = \mathbf{p}^i \quad (1)$$

Where, $\mathbf{x} \in \mathbb{R}^{3n}$ a column vector concatenating x, y, z coordinates from a set of n vertices $\mathbf{v}_1, \dots, \mathbf{v}_n$ of the target mesh. $\mathbf{A} \in \mathbb{R}^{9F \times 3n}$ is a sparse matrix whose elements of a diagonal submatrix $\in \mathbb{R}^{3F \times n}$ come from a QR decomposition of the target deformation gradients $\mathbf{D}^{\tilde{\mathcal{T}}}$, and $\mathbf{B} \in \mathbb{R}^{9F}$ is a column vector stacking x, y, z element of deformation gradients for all faces from the source mesh animation. Clearly, there are no explicit constraints to avoid self-intersections.

2.2 Self-intersection awareness

To suppress self-intersections on the resulting mesh from DT, we minimize the self-intersection energy similar to the sliding constraint proposed by Li et al [Li and Barbič 2019], based on the coplanarity conditions between a point and a triangle

$$E_{sint} = \|\mathbf{C}\mathbf{x} - \mathbf{c}\|_2^2 \quad (2)$$

where r -th row of the matrix $\mathbf{C} \in \mathbb{R}^{|r| \times 3n}$ and $\mathbf{c} \in \mathbb{R}^{|r| \times 1}$ are as following.

$$\mathbf{C}_{r,:} = [\mathbf{n}_{i,x}S, \mathbf{n}_{i,y}S, \mathbf{n}_{i,z}S], \quad \mathbf{c}_r = w d_{bb} \|\mathbf{n}_i\|^2 \quad (3)$$

where $\mathbf{n}_i$ is a normal vector of the i -th triangle colliding with the point and $S = (\mathbf{s}_i - \sum_{j \in f} \lambda_j \mathbf{s}_j)^\top \in \mathbb{R}^{1 \times n}$. $\mathbf{s}_i \in \mathbb{R}^n$ is a selection vector whose elements are 1 only for the i -th vertex. $\mathbf{s}_j$ is also a selection vector whose element is 1 only for the j -th vertex on the f -th triangle. λ_j is a barycentric coordinate. $|r|$ is the number of detected self-intersections. d_{bb} is a diagonal length of the bounding box of the target mesh in rest pose. w is a weight for controlling minimum separation between a vertex and a contact triangle. We empirically set $w=0.03$ which corresponds to 3% of d_{bb} . For continuous collision detection, we developed our own CCD library based on [Tang et al. 2009].

2.3 Temporal Smoothness

To ensure temporal consistency of the resulting mesh sequence, we minimize the acceleration energy for all vertices $\mathbf{x}$ over the mesh at time t :

$$E_{tempo} = \|\tilde{\mathbf{x}}_{t+1} - 2\mathbf{x} + \mathbf{x}_{t-1}\|_2^2 \quad (4)$$

Where, $\tilde{\mathbf{x}}_{t+1}$ comes from the mesh vertices deformed by original DT at next frame and $\mathbf{x}_{t-1}$ is from SIDT at last frame.

2.4 Optimization

By combining all energy terms described above, we can form the SIDT system that enables us to obtain a deformed mesh pose by minimizing the following total energy with respect to $\mathbf{x}$.

$$\min_{\mathbf{x}} E_{dt} + \alpha E_{sint} + \beta E_{tempo} \quad \text{s.t.} \quad \mathbf{x}_k = \mathbf{p}^i \quad (5)$$

Where, α and β are weights for each energy term, and we empirically set $\alpha = 1.0, \beta = 0.01$. The optimization runs if self-intesections occur on the resulting mesh after original DT only. Intersections may not be completely resolved by a single iteration. In such a case, we repeatedly solve Eq. 5 until intersections are totally eliminated or we otherwise reach maximum number of iterations.

3 CASE STUDY AND DISCUSSIONS

To demonstrate the usefulness of SIDT, we have applied it to a garment simulation pipeline. Our pipeline starts with creating a mesh animation in sport motion for a subject, then we add cloth mimicking the material properties of the prototype sportswear, finally we compute strain simulation. A subject's mesh model is acquired by using commercial 3D whole body scanner. To create a subject's mesh animation, we use DT or SIDT to transfer deformations of a rigged template human mesh model to the subject's mesh model. The template human model can be animated by MoCap. In this paper, we used motion capture library of freestyle swimming and running. For comparison, we apply *Animation Repair*¹ as a post-process to the resulting mesh animation from DT to solve self-intersections.

Results are shown in Fig. 1 and the supplemental video. In the case of DT, garment simulation failed in the middle of the sequence due to the self-intersection of the right hand and the waist (Fig. 1(b)). In this case, we need to manually fix the model by manipulating rig of the template model with tedious trial-and-error (Fig. 1(e)), and it took 10 days for a CAE operator who is not familiar with CG animation work. *Animation Repair* helps to resolve intersections, however requires more than 1 hour and may generate some distortions in the result mesh depending on the $\hat{d}$. In contrast, SIDT can address this issue in about 15 minutes, without any manual interventions. The garment simulation can complete without generating artifacts (Fig. 1(d)). When we compare the strain distribution of cloth on manually modified model (Fig. 1(e)) and SIDT, they have similar appearances. However, temporal jitter and varying volume were slightly observed in our experiment. These did not affect our garment simulation but need to be solved for further improvements.

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¹We set $\hat{d} = 0.003 \sim 0.01$ meters. It means distance at which contact repulsions can work [Fang et al. 2021]. Other settings are followed by the paper.